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Reactive Spark Plasma Sintering of the hexacelcian barium aluminosilicate phase.

C. Guerin¹, A. Bassoulet-Saintonge¹, A. Allemand^{1,2}, A. Maitre³, R. Boulesteix³, T. Piquero², and Y. Le Petitcorps¹

1) CNRS, LCTS, UMR 5801, F-33600 Pessac, France

2) CEA, DAM, LE RIPAUT, F-37260 Monts, France

3) CNRS, IRCER, UMR 7315, F-87068 Limoges, France

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Abstract

The barium aluminosilicate compound ($\text{BaAl}_2\text{Si}_2\text{O}_8$ or BAS) has been synthesized by powder reactive sintering within a Spark Plasma Sintering (SPS) device. The reaction paths between the precursors (alumina, silica and barium carbonate powders) have been deeply investigated at different temperatures from 900 to 1550°C in order to get at the end, the hexacelcian crystallographic form without any undesirable other compounds. Thanks to this approach, a 3 step thermal treatment is proposed to get a fully dense and nearly pure (98 wt%) BAS.

Introduction

The synthesis of BAS ($\text{BaAl}_2\text{Si}_2\text{O}_8$) has been widely studied for its physical properties (low dielectric permittivity) as a bulk material or as a matrix in composite materials reinforced with silica, alumina or silicon carbide fibers [1]–[11]. The BAS is a line compound with a congruent melting temperature 1750 °C, it has been firstly studied by Toropov et al. [12], [13].

Like most of the oxide compounds, the BAS exists under various crystallographic structures depending on the temperatures: monoclinic (stable from room temperature up to 1590°C) and hexacelsian or hexagonal (stable between 1590°C and its melting point) [14], [15]. The monoclinic phase has a lower coefficient of thermal expansion compared to the hexagonal one ($2.29.10^{-6}\text{°C}^{-1}$ / 8.10^{-6}°C^{-1} respectively) [16], [17]. Therefore, numerous studies were focused on the monoclinic phase for its physical properties on the wide range of temperature. However, some specific applications at higher temperatures (between 1600 and 1700°C) may require the use of the hexagonal phase, for example in the field of composite materials where the BAS matrix is reinforced by alumina fibers. [18].

Numerous routes for the synthesis of BAS have been investigated: reactive sintering [19], [20], sol-gel [21], [22], metallic or zeolite precursors [23], [24], or direct melting [6], [25], [26]. In the solid-state process involving the reactive sintering, three precursors in a fine (1-5 μ m) powder shape are commonly used: the calcium carbonate, the alumina and silica ones. Different heating devices and sintering steps are employed like: (1) a conventional furnace, (2) the Spark Plasma Sintering (SPS) or (3) the Ultra-fast Pressureless Sintering (UFPS) [20]. In a conventional furnace, under atmospheric air pressure, Planz et al. showed that the BAS is formed since 1000°C [27]. Several studies have been done between 1000°C to 1500°C for durations as long as 160 hours in order to get monoclinic or a mixture of hexagonal and monoclinic phases [28]–[30]. Lee [31] studied the synthesis of BAS in a conventional furnace and shows that it follows a two step mechanisms : (1) A reactive sintering at 1150°C during 4 hours between two precursors (BaO-SiO₂ or BaO-Al₂O₃) to form respectively a barium silicate or barium aluminate intermediate compound and (2) this intermediate compound reacts with the remaining precursor at 1200°C during 6 hours to form the BAS. More recently, Billard evaluated this route, involving an aluminate intermediate to get nearly 90 wt% of the hexagonal phase and 10 wt% of residual barium aluminate and alumina phases [19]. The reaction paths have been also well described by Lee and Aswath [23], [32]. Using the Spark Plasma Sintering (SPS) device, Ye et al. fabricate a BAS matrix for preparing composite materials [19]. Three composite systems were investigated: BAS/Si₃N₄ [33], BAS/ α -SiAlON [34] and BAS/SiC [35].

The objective of this study is to sinter pure hexagonal BAS by SPS. The SPS is an ultra-fast sintering process (100°C/min) and the sintering duration can be short (5-30 min). In order to have a fully reaction between the 3 starting powders (alumina, silica, barium carbonate), it is very important to understand the pathway of the various chemical reactions which can occurred. Knowing that, intermediate thermal treatments are evaluated in order to get at the end, the hexagonal BAS completely sintered, fully dense and pure.

Materials and Methods

The starting powders were: BaCO₃ (99%wt grade, Sigma Aldrich, d₅₀=3.4 μ m), SiO₂ quartz (99%wt grade, Sigma Aldrich, d₅₀=1.4 μ m) and α -Al₂O₃ (99%wt grade, Sigma Aldrich, d₅₀=0.2 μ m). The powders were weighted in stoichiometric proportions (2:1:1 respectively for SiO₂, Al₂O₃ and BaCO₃). The powders were together water blended and the mixture was then homogenized by a sonicator (Vibrocell 75, 401) during 3 minutes and milled using a planetary grinder (Pulverisette 7 premium line) with alumina balls (Φ =5mm in diameter) according to 5 cycles (2 min On/ 1 min Off) at 300 rpm. The suspension was then dried at 100°C during 4h.

Then the dried powders were poured into a graphite mold. A layer of boron nitride was sprayed onto the graphite surfaces to reduce the carbon contamination. A carbon felt surrounds the graphite mold to guarantee an homogeneous temperature inside the sample and to protect the SPS

chamber against the risk of the rupture of the graphite mold during pressing. The SPS device used for this study was the SPS-511S model by Dr.Sinter, allowing the sintering of cylindrical samples (10mm in diameter and 5mm in height for this study). In order to reach quickly the sintering temperature, the heating rate applied was 100 °C/min, the atmosphere within the SPS chamber is vacuum. For this study, interrupted heat treatments were carried out. To do that, the SPS device was stopped at the temperature required and the temperature decreased according to the apparatus inertia.

For the characterizations, the open porosity and density were measured according to the Archimedes method. The room temperature XRD was carried out using Cu K α radiation in a Bruker D8 Advance system. The power supply was set to 40kV – 40mA with a step size of 0.01 (2 θ), with a scan rate of 2 seconds per step. Each analysis was done in the range of 10 to 80° (2 θ angle) in order to identify all the compounds. Finally, the EVA software [36] was used to identify and quantify the proportion of the different phases. A morphological and chemical study of the BAS was also done using a scanning electron microscope (SEM (Quanta-FEI)) coupled with an Energy Dispersion Spectroscopy (EDS) and Electron Back Scattered Diffusion (EBSD) to determine the grain size morphology and composition.

Results and discussions

Preliminary Reactive Sintering: incomplete reaction

A first thermal treatment was performed at 1200 °C for 10 minutes. As soon as the temperature reaches 1200°C, the material melt. However Lee and Aswath have explained that the reactions leading to the hexacelsian formation occur since 1050 °C [23]. Before this step, various intermediate compounds like barium silicate or aluminate may be present. The liquidus projection of the ternary phase diagram (BaO-SiO₂-Al₂O₃), shows that barium aluminosilicate (BAS) and the barium silicate present few liquid eutectic compositions in this range of temperatures (between 1200 and 1400 °C) [12], [13].

During this thermal treatment by SPS, a high heating rate is used. This rate is too fast to allow the complete barium silicate reaction with alumina to form the BAS, consequently, the eutectic composition was reached and the material melts.

To avoid this, intermediate plateaus in the thermal cycle are necessary to allow a complete reaction. Thanks to these plateaus, a majority of the barium silicates may react, avoiding the melting of the material.

Thermal cycle adjustments

According to the study done by Lee and Aswath, the reactional path followed three different steps [23] : (1) BaCO₃ complete reaction with silica and alumina powders, (2) hexacelsian

precipitation through the silicate path and/or (3) hexacelsian formation through the aluminate path. To allow this, the SPS thermal cycle needed to be adjusted as follow:

(1) BaCO₃ reaction with silica

When the temperature reaches 700 °C to 900 °C, there is a great increase of the pressure inside the SPS chamber. Then, during the plateau at 900 °C, the pressure decreased (**figure 1**). To understand the reactive sintering steps and the various possible chemical reactions, four interrupted thermal treatments were carried out and the samples were then analyzed by XRD (**figures 2-3** and **table 1**).

At 600 °C (I), only the three primary powders are present: BaCO₃ (W), SiO₂ (S) and Al₂O₃ (A). At 900 °C (II), it can be noticed a slight decrease of BaCO₃ and BaSiO₃ (BS) is formed (**table 1**). The system began to react; this was the low temperature step in the system evolution, that is to say the reaction between barium carbonate W and silica S to form barium silicate BS. The increase of the pressure in the SPS chamber was due to the CO₂ release from this reaction (**figure 1**). It can be seen also from this figure that the degassing was not over at the end of the one-minute plateau at 900 °C. The XRD analysis confirmed that the duration of the plateau was not long enough to allow the carbonate to react completely. A 30-minute plateau is however enough and almost all the BaCO₃ phase was consumed by the reaction with the silica. As a consequence, the gas pressure in the SPS chamber is constant for this thermal treatment (**figure 4**). At the end of the 30-minute plateau at 900 °C, the system did not degas anymore. The first step of the reaction is now finished. This plateau will be maintained in the final thermal cycle for the hexagonal BAS elaboration by SPS.

(2) Hexacelsian precipitation through the silicate route

At 1200 °C the material could melt due to the eutectic compositions between the barium silicate and barium aluminosilicate (BAS). To limit this risk, an intermediate plateau below 1200°C is required. To determine the temperature and the duration of this plateau, new interrupted heat treatments were carried out. The compositions of the reacted products were checked by XRD (**figures 2-3** and **table 1**).

When the temperature is in between 900 °C to 1150 °C (IV and V), the beginning of the aluminate path reaction is explained by the barium aluminate formation. Moreover, the hexacelsian precipitation and the W and BS disappearance indicated that the silicate path was still progressing. In the silicate path described by Lee and Aswath [23], W and BS were the first reagents, then the sequence of precipitations is respectively : B₂S₃ and B₃S₅, and lastly BS₂. In this study, B₃S₅ was formed during the temperature increase. Even if, the SPS provides a high heating rate, the system could react. After 10 minutes at 1150 °C (VI), the amount of BA was constant. The H formation by the aluminate path did not occur anymore. Nevertheless, the silicate path progressed, no more B₃S₅ was present in the material, and the last barium silicate intermediate BS₂ was formed, at the end of this step

the amount of H starts to increase. After 30 minutes at 1150 °C (VII), the system did not react anymore.

Thanks to these experiments, the second plateau at 1150 °C was set to 10 minutes of duration. This plateau allows to initiate the aluminate path and to favor the silicate path progression. By this way, the amount of barium silicate in the system is reduced thus limiting the melting risk at higher temperatures.

(3) Hexacelsian precipitation through the aluminate path

Interrupted heat treatments at 1400°C were then carried out to follow the reactions (**figures 3, 5 and table 1**).

The silicate path reaction continues during this treatment at 1400 °C, this is confirmed by the XRD results (samples VII and VIII, **figure 3**). However, no more barium silicate intermediate compounds were noticeable now. Nevertheless, a large quantity of alumina was still present. As Lee and al. presented in their study [23], the system continues to react even if, only two phases (H and A) are detected by XRD. An enriched phase in silicon and barium, which is not detectable by XRD analysis (probably amorphous or in a small quantity), is still present and reacts with the remaining alumina to form hexagonal BAS. SEM analyses detected this phase. This is confirmed by the decrease in BA and the last reaction $BA + S \rightarrow H$ occurred. After 10 minutes at 1400 °C (IX), the aluminate path was finished. The entire BA was consumed, the amount of alumina decreased.

To reduce the amount of impurities, a last treatment at 1400 °C for 60 minutes (X) was carried out. The XRD results (**Table 1**) showed that the amount of alumina decreased and the amount of hexacelsian H increased to 92 wt%.

To complete the reaction, a last treatment at 1550 °C for 60 minutes (XI) was carried out. XRD analysis shows that the sample contains 98 wt% of hexacelsian and only 2 wt% of alumina (**Table 1**).

The relative density measured by Archimedes method is 98%. The SEM microstructures showed a small amount of closed porosity (**figure 6**). EDS analysis showed that the material was mainly composed of BAS. Few precipitates of alumina and a phase enriched in silicon and barium were observed. The grain size distribution and the grain composition of this sample have been quantified by an EBSD analysis (**figure 7**). The mean grain diameter was about 0.95 µm with a standard deviation of 0.43 µm, this grain size is relatively small which may be interesting for mechanical applications.

Conclusions and prospective

In order to synthesize the hexacelcian barium aluminosilicate compound ($\text{BaAl}_2\text{Si}_2\text{O}_8$ or BAS) by reactive spark plasma sintering of alumina, silica and barium carbonate powders various thermal treatments (in temperatures and durations) were investigated. Knowing the reaction path and the mechanism between the reactants, a 3 step thermal cycle has been proposed. The first plateau at 900 °C during 30 minutes allows the reaction of the barium carbonate with the silica to form barium silicates intermediates. The second one at 1150 °C for 10 minutes corresponds to the silicates reaction, the hexacelsian precipitation starts from the silicate path and the barium aluminate occurs. And the last plateau at 1550 °C for 60 minutes allows the hexcelsian formation from the silicate and the aluminate path. Applying these thermal cycles, a fully dense and nearly pure, small grain size hexacelcian BAS is obtained.

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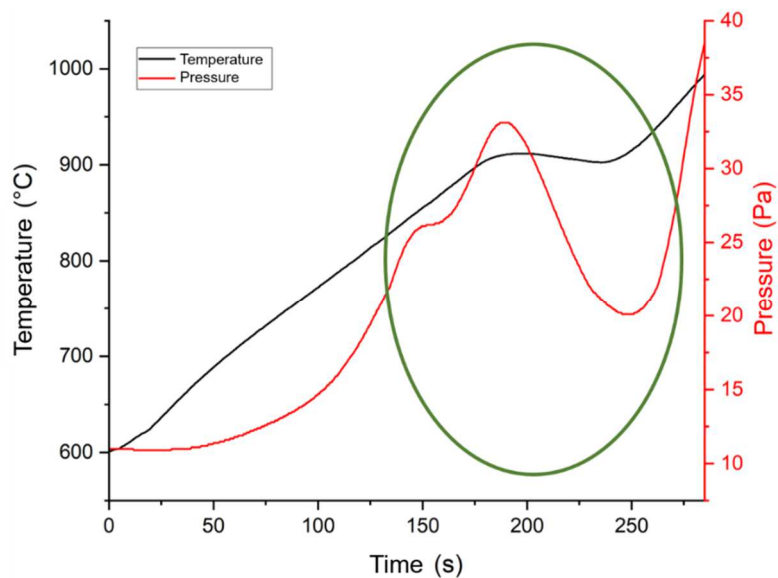


Figure 1: Temperature and pressure versus time inside the SPS, the increase of the pressure is due to the decomposition of the calcium carbonate (CO_2 release), the reaction is not complete after 300s.

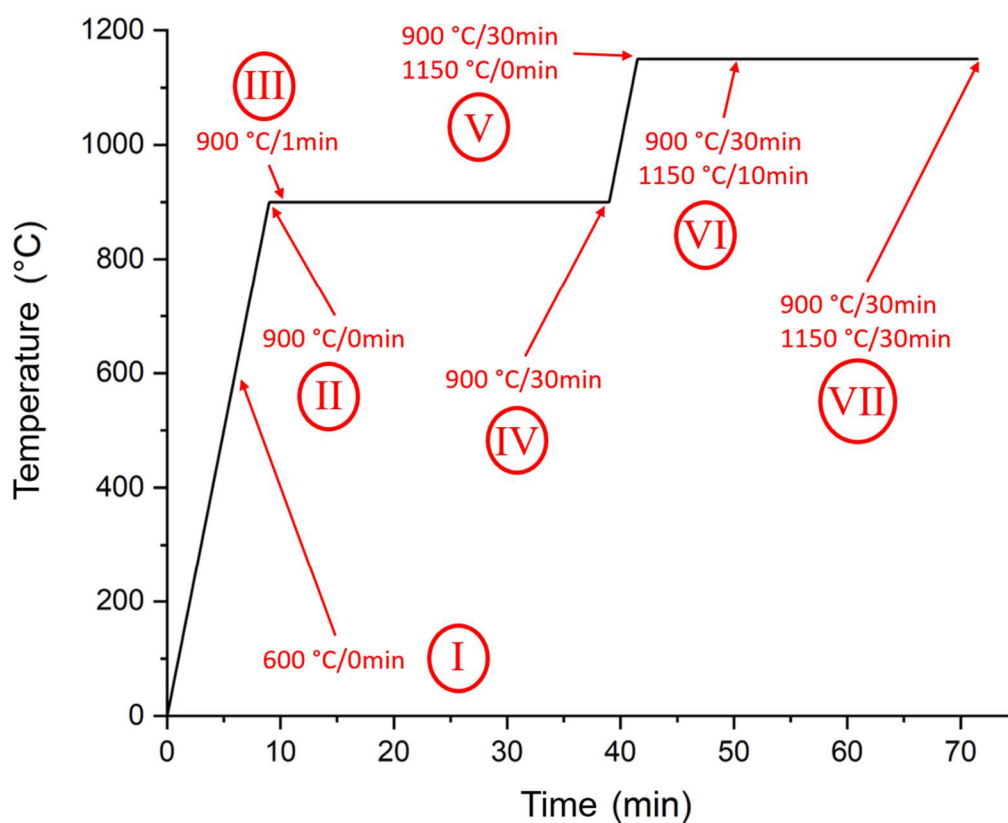


Figure 2: Thermal treatments (I to VII) realized in order to study the different reactions between BaCO_3 , SiO_2 and Al_2O_3 and their compounds

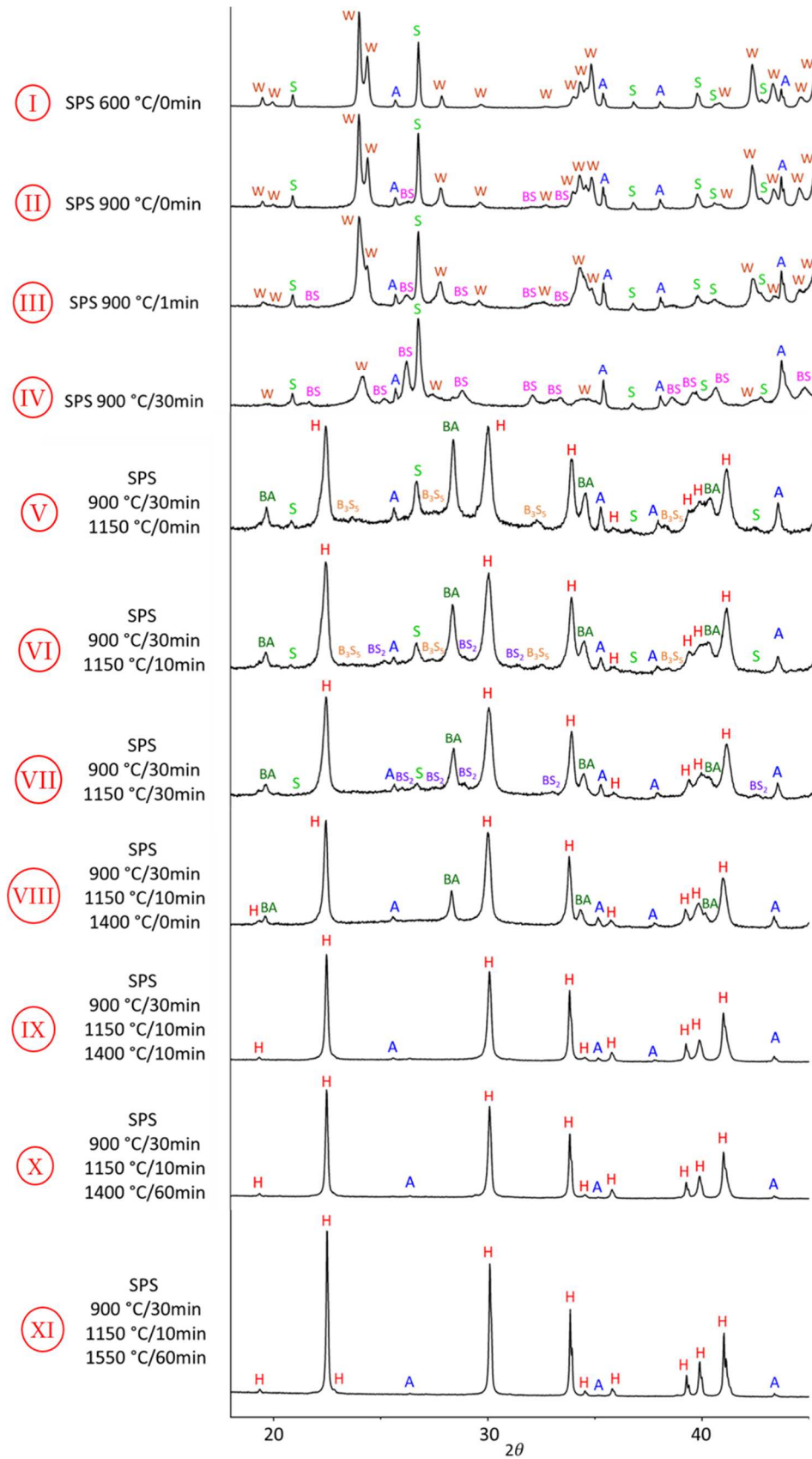


Figure 3: Room temperature XRay Diffraction patterns of samples corresponding to the different thermal treatments (samples I to XI), the identification (H, A,BA,..) are given in table 1.

Sample	Compounds Thermal treatment sequence	W	A	S	BS	B ₃ S ₅	BS ₂	BA	H
		BaCO ₃ (Wt.%)	Al ₂ O ₃ (Wt.%)	SiO ₂ (Wt.%)	BaSiO ₃ (Wt.%)	Ba ₃ Si ₅ O ₁₃ (Wt.%)	BaSi ₂ O ₅ (Wt.%)	BaAl ₂ O ₄ (Wt.%)	BaAl ₂ Si ₂ O ₈ (Wt.%)
I	600 °C / 0 min	25	Bal.	Bal.	-	-	-	-	-
II	900 °C / 0 min	24	Bal.	Bal.	2	-	-	-	-
III	900 °C / 1min	23	Bal.	Bal.	4	-	-	-	-
IV	900 °C / 30min	5	Bal.	Bal.	17	-	-	-	-
V	900 °C/30min 1150 °C	-	39	11	-	10	-	20	20
VI	900 °C/30min 1150 °C/10min	-	30	4	-	-	14	21	31
VII	900 °C/30min 1150 °C/30min	-	34	4	-	-	14	17	31
VIII	900 °C/30min 1150 °C/10min 1400 °C/0min	-	35	-	-	-	-	18	47
IX	900 °C/30min 1150 °C/10min 1400 °C/10min	-	22	-	-	-	-	-	78
X	900 °C/30min 1150 °C/10min 1400 °C/60min	-	8	-	-	-	-	-	92
XI	900 °C/30min 1150 °C/10min 1550 °C/60min	-	2	-	-	-	-	-	98

Table 1: XRD semi-quantification of the compounds present in the samples I to XI

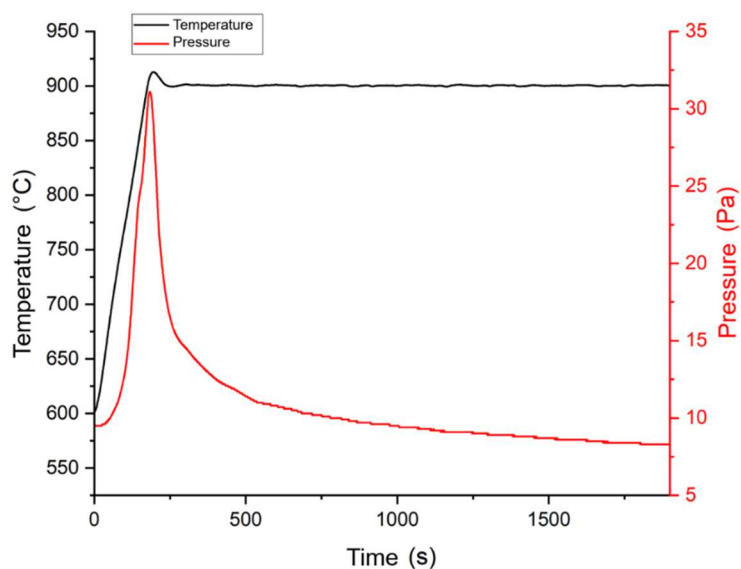


Figure 4: Temperature and gas pressure versus time inside the SPS chamber, after 500 s the reaction with BaCO₃ is completed.

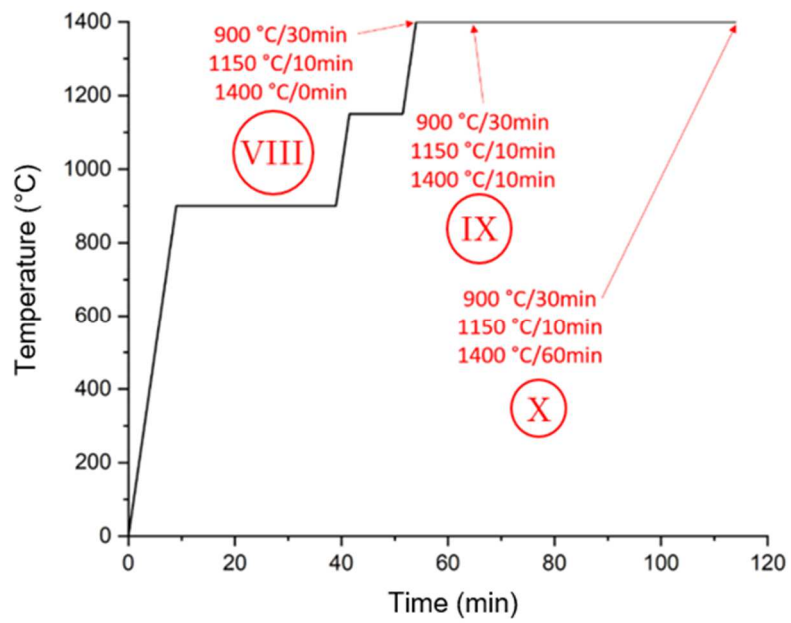


Figure 5: Last interrupted thermal treatments (VIII to X) in order to get a pure Hexacelcian composition

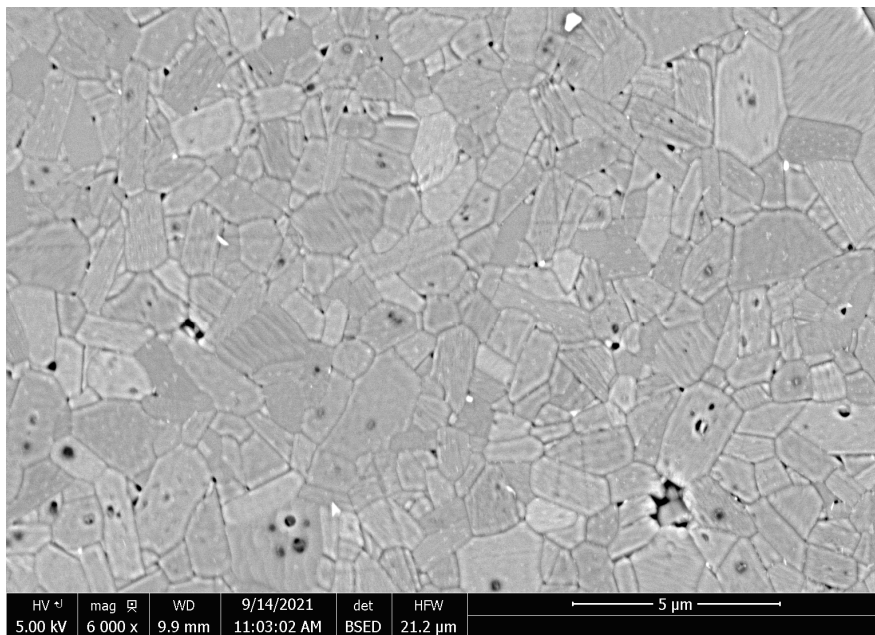


Figure 6: SEM microstructures of a sample sintered at 1550 °C during 60 min. The material has been heat treated 30min at 1100°C to reveal its microstructure.

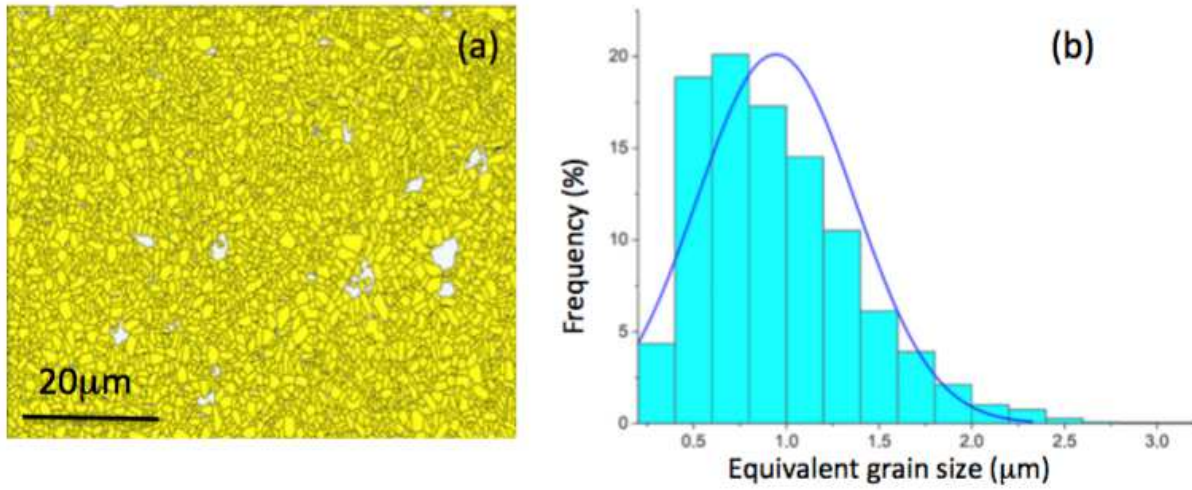


Figure 7: EBSD analysis : (a) mapping of the grains and (b) Grain size distribution. The BAS has been sintered at 1550 °C for 60 min by SPS

References

- [1] A. Bandyopadhyay, P. B. Aswath, W. D. Porter, et O. B. Cavin, « The low temperature hexagonal to orthorhombic transformation in Si₃N₄ reinforced BAS matrix composites », *J. Mater. Res.*, vol. 10, n° 5, p. 1256-1263, 1995.
- [2] Y. S. Touloukian, R. K. Kirby, E. R. Taylor, et T. Y. R. Lee, « Thermophysical Properties of Matter - the TPRC Data Series. Volume 13. Thermal Expansion - Nonmetallic Solids », Thermophysical and electronic properties information analysis center Lafayette in, 1977
- [3] R. Mccauley, « Polymorphism and Dielectric Electric Properties of Ba- and Sr-Containing Feldspars », *J. Mater. Sci.*, vol. 35, p. 3939-3942, 2000, doi: 10.1023/A:1004802002579.
- [4] J. A. Zaykoski et I. G. Talmy, « Gallium and Germanium Substitutions in Celsian », in *Proceedings of the 18th Annual Conference on Composites and Advanced Ceramic Materials—B: Ceramic Engineering and Science Proceedings*, John Wiley & Sons, Ltd, 1994, p. 778-786. doi: 10.1002/9780470314555.ch23.
- [5] X.-D. Zhang, K. H. Sandhage, et H. L. Fraser, « Synthesis of BaAl₂Si₂O₈ from Solid Ba-Al-Al₂O₃-SiO₂ Precursors: II, TEM Analyses of Phase Evolution », *J. Am. Ceram. Soc.*, vol. 81, n° 11, p. 2983-2997, 2005, doi: 10.1111/j.1151-2916.1998.tb02724.x.
- [6] N. P. Bansal, M. J. Hyatt, et C. H. Drummond, « Crystallization and Properties of Sr-Ba Aluminosilicate Glass-Ceramic Matrices », in *Ceramic Engineering and Science Proceedings*, vol. 12, J. B. Wachtman, Éd. Hoboken, NJ, USA: John Wiley & Sons, Inc., 1991, p. 1222-1234. doi: 10.1002/9780470313831.ch23.
- [7] M. Starczewski, « Resistance of Celsian to Some Corrosive Agents », *Pr Inst Hutn*, n° 14, p. 69-74, 1962.
- [8] H. C. Lin et W. R. Foster, « Studies in the System BaO-Al₂O₃-SiO₂: V. The Ternary System Sanbornite-Celsian-Silica * », *J. Am. Ceram. Soc.*, vol. 53, n° 10, p. 549-551, 1970, doi: 10.1111/j.1151-2916.1970.tb15963.x.
- [9] J. M. Wright, J. F. Meyers, et E. E. Ritchie, « NASA Conference Publication No. 3175, 16th Conference on Metal Matrix, Carbon, and Ceramic Matrix Composites », *Natl. Aeronaut. Space Adm.*, vol. Part. 2, p. 845-854, 1992.
- [10] V. S. R. Murthy et M. H. Lewis, « Matrix crystallization and interface structure in SiC–celsian composites », *Br Ceram Trans J*, vol. 89, n° 5, p. 173-174, 1990.
- [11] K. N. Lee *et al.*, « Upper Temperature Limit of Environmental Barrier Coatings Based on Mullite and BSAS », *J. Am. Ceram. Soc.*, vol. 86, n° 8, p. 1299-1306, 2003, doi: 10.1111/j.1151-2916.2003.tb03466.x.
- [12] N. A. Toropov, F. Ya. Galakhov, et I. A. Bondar, « The diagram of state of the ternary system BaO-Al₂O₃-SiO₂ », *Bull. Acad. Sci. USSR Div. Chem. Sci.*, vol. 3, n° 5, p. 647-655, 1954, doi: 10.1007/BF01170102.

- [13] C. E. Semler et W. R. Foster, « Studies in the System BaO-Al₂O₃-SiO₂ : VI, The System Celsian-Silica-Alumina », *J. Am. Ceram. Soc.*, vol. 53, n° 11, p. 4, 1970.
- [14] B. Yoshiki et K. Matsumoto, « High-Temperature Modification of Barium Feldspar », *J. Am. Ceram. Soc.*, vol. 34, n° 9, p. 283-286, 1951.
- [15] H. C. Lin et W. R. Foster, « Studies in the system BaO-Al₂O₃-SiO₂ I. The polymorphism of celsian », *Am. Mineral.*, vol. 53, n° 1-2, p. 134-144, 1968.
- [16] Y. Kobayashi, « Transformation kinetics from hexacelsian to celsian for powders having uniform particle size », *Ceram. Int. - CERAM INT*, vol. 27, p. 179-184, 2001, doi: 10.1016/S0272-8842(00)00059-6.
- [17] N. Fréty, A. Taylor, et M. H. Lewis, « Microstructure and crystallisation behaviour of sol-gel derived 1/2SrO□1/2BaO□Al₂O₃□2SiO₂ glass-ceramic », *J. Non-Cryst. Solids*, vol. 195, n° 1, p. 28-37, 1996, doi: 10.1016/0022-3093(95)00519-6.
- [18] A. Allemand, P. Belleville, Y. L. Petitcorps, R. Billard, et M. Dolle, « Method for preparing a material made from aluminosilicate and method for preparing a composite material having an aluminosilicate matrix », US10717656B2, 2020
- [19] R. Billard, « Mise au point d'un composite à fibre oxyde et matrice d'aluminosilicate de baryum modifiée », Université de Bordeaux, 2015.
- [20] A. Allemand, C. Guerin, C. Besnard, R. Billard, et Y. Le Petitcorps, « A comparison between a new Ultra Fast Pressureless Sintering (UFPS) technology and Spark Plasma Sintering (SPS) for Barium AluminoSilicate metastable phase », *J. Eur. Ceram. Soc.*, vol. 41, n° 2, p. 1524-1529, 2021, doi: 10.1016/j.jeurceramsoc.2020.09.054.
- [21] F. Ye, J. C. Gu, Y. Zhou, et M. Iwasa, « Synthesis of BaAl₂Si₂O₈ glass-ceramic by a sol-gel method and the fabrication of SiCp/BaAl₂Si₂O₈ composites », *J. Eur. Ceram. Soc.*, vol. 23, n° 13, p. 2203-2209, 2003.
- [22] L. Zhien, H. Ying, et Y. Jianjun, « Study of the preparation process for BaO-Al₂O₃-SiO₂ powders by a two-step method », *J Mater Sci*, vol. 30, p. 405-409, 1995.
- [23] K.-T. Lee et P. B. Aswath, « Synthesis of Hexacelsian Barium Aluminosilicate by a Solid-State Process », *J. Am. Ceram. Soc.*, vol. 83, n° 12, p. 2907-2912, 2000, doi: 10.1111/j.1151-2916.2000.tb01659.x.
- [24] B. Hoghooghi, J. Mckittrick, E. Helsel, et O. Lopez, « Microstructural Development, Densification, and Hot Pressing of Celsian Ceramics from Ion-Exchanged Zeolite Precursors », *J. Am. Ceram. Soc.*, vol. 81, p. 845-852, 2005, doi: 10.1111/j.1151-2916.1998.tb02418.x.
- [25] W. E. Lee, M. Chen, et P. F. James, « Crystallization of Celsian (BaAl₂Si₂O₈) Glass », *J. Am. Ceram. Soc.*, vol. 78, n° 8, p. 2180-2186, 1995.
- [26] D. Bahat, « Kinetic study on the hexacelsian-celsian phase transformation », *J. Mater. Sci.*, vol. 5, n° 9, p. 805-810, 1970.

- [27]J. E. Planz et H. Müller-Hesse, « Solid-state reactions in the system BaO-Al₂O₃-SiO₂. II. The formation of celsian by solid-state reactions », *Berichte Dtsch. Keram. Ges.*, vol. 40, n° 3, p. 191-200, 1963.
- [28]M. M. Krzmanc, M. Valant, et D. Suvorov, « The synthesis and microwave dielectric properties of Sr_xBa_{1-x}Al₂Si₂O₈ and Ca_yBa_{1-y}Al₂Si₂O₈ ceramics », *J. Eur. Ceram. Soc.*, vol. 27, n° 2, p. 1181-1185, 2007, doi: 10.1016/j.jeurceramsoc.2006.05.017.
- [29]I. G. Talmy et D. A. Haught, « Celsian-based (BaO.Al₂O₃.SiO₂) ceramics as candidates for radomes », 1990, vol. 3097, p. 239-250.
- [30]Y. Tabira, R. Withers, Y. Takéuchi, et F. Marumo, « Structured diffuse scattering, displacive flexibility and polymorphism in Ba-hexacelsian », *Phys. Chem. Miner.*, vol. 27, p. 194-202, 2000, doi: 10.1007/s002690050007.
- [31]K.-T. Lee, « Studies of hexacelsian and celsian barium aluminosilicates », 1998
- [32]K.-T. Lee et P. B. Aswath, « Kinetics of the hexacelsian to celsian transformation in barium aluminosilicates doped with CaO », *Int. J. Inorg. Mater.*, vol. 3, n° 7, p. 687-692, 2001, doi: 10.1016/S1466-6049(01)00190-8.
- [33]F. Ye, L. Liu, J. Zhang, et Q. Meng, « Synthesis of 30wt%BAS/Si₃N₄ composite by spark plasma sintering », *Compos. Sci. Technol.*, vol. 68, n° 3-4, p. 1073-1079, 2008, doi: 10.1016/j.compscitech.2007.07.001.
- [34]F. Ye, L. Zhang, C. Liu, limeng Liu, et H. Zhang, « Thermal shock resistance of in situ toughened α -SiAlONs with barium aluminosilicate as an additive sintered by SPS », *Mater. Sci. Eng. A*, vol. 527, n° 23, p. 6368-6371, 2010, doi: 10.1016/j.msea.2010.06.074.
- [35]J. Ma, F. Ye, et Z. Hua, « Microstructure and mechanical properties of BAS/SiC composites sintered by spark plasma sintering », *Mater. Charact.*, vol. 84, p. 100-104, 2013, doi: 10.1016/j.matchar.2013.06.014.
- [36]H. M. Rietveld, « A profile refinement method for nuclear and magnetic structures », *J. Appl. Crystallogr.*, vol. 2, n° 2, p. 65-71, 1969, doi: 10.1107/S0021889869006558.